STACKS, COHOMOLOGY OF SCHEMES (COHERENT.TEX): LEMMA-COHOMOLOGY-PROJECTIVE-SPACE-OVER-RING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1819-1839 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 1841-2057, the complete author proof (215 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_coherent.json (pb=7679, stmt 1819-1839, proof 1841-2057).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01FM: sources/stacks-coherent/pinned-original-source.tex, source lines 2057–2057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01FJ: sources/stacks-coherent/pinned-original-source.tex, source lines 2057–2057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01FG: sources/stacks-coherent/pinned-original-source.tex, source lines 2057–2057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01M3: sources/stacks-coherent/pinned-original-source.tex, source lines 2057–2057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XD: sources/stacks-coherent/pinned-original-source.tex, source lines 463–473. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01X9: sources/stacks-coherent/pinned-original-source.tex, source lines 262–269. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let R be a ring. Let $n \geq 0$ be an integer. We have*

$$H^q(\mathbf{P}^n, \mathcal{O}_{\mathbf{P}^n_R}(d)) = \begin{cases} (R[T_0, \ldots, T_n])_d & \text{if} \quad q = 0,\ d \geq 0 \\ \left(\frac{1}{T_0 \ldots T_n} R[\frac{1}{T_0}, \ldots, \frac{1}{T_n}]\right)_d & \text{if} \quad q = n,\ d < 0 \\ 0 & \text{if} \quad q, n, d \text{ not as above} \end{cases}$$

as R-modules.

Proof. We will use the standard affine open covering

$$\mathcal{U} : \mathbf{P}^n_R = \bigcup\nolimits_{i=0}^{n} D_+(T_i)$$

to compute the cohomology using the Čech complex. This is permissible by Lemma 01XD since any intersection of finitely many affine $D_+(T_i)$ is also a standard affine open (see Constructions, Section 01M3). In fact, we can use the alternating or ordered Čech complex according to Cohomology, Lemmas 01FJ and 01FM.

The ordering we will use on $\{0, \ldots, n\}$ is the usual one. Hence the complex we are looking at has terms

$$\check{\mathcal{C}}^p_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d)) = \bigoplus\nolimits_{i_0 < \ldots < i_p} (R[T_0, \ldots, T_n, \frac{1}{T_{i_0} \ldots T_{i_p}}])_d$$

Moreover, the maps are given by the usual formula

$$d(s)_{i_0 \ldots i_{p+1}} = \sum\nolimits_{j=0}^{p+1} (-1)^j s_{i_0 \ldots \hat{i}_j \ldots i_{p+1}}$$

see Cohomology, Section 01FG. Note that each term of this complex has a natural $\mathbf{Z}^{n+1}$-grading. Namely, we get this by declaring a monomial $T_0^{e_0} \ldots T_n^{e_n}$ to be homogeneous with weight $(e_0, \ldots, e_n) \in \mathbf{Z}^{n+1}$. It is clear that the differential given above respects the grading. In a formula we have

$$\check{\mathcal{C}}^\bullet_{ord}(\mathcal{U}, \mathcal{O}_{\mathbf{P}_R}(d)) = \bigoplus\nolimits_{\vec{e} \in \mathbf{Z}^{n+1}} \check{\mathcal{C}}^\bullet(\vec{e})$$

where not all summands on the right hand side occur (see below). Hence in order to compute the cohomology modules of the complex it suffices to compute the cohomology of the graded pieces and take the direct sum at the end.

Fix $\vec{e} = (e_0, \ldots, e_n) \in \mathbf{Z}^{n+1}$. In order for this weight to occur in the complex above we need to assume $e_0 + \ldots + e_n = d$ (if not then it occurs for a different twist of the structure sheaf of course). Assuming this, set

$$NEG(\vec{e}) = \{i \in \{0, \ldots, n\} \mid e_i < 0\}.$$

With this notation the weight $\vec{e}$ summand $\check{\mathcal{C}}^\bullet(\vec{e})$ of the Čech complex above has the following terms

$$\check{\mathcal{C}}^p(\vec{e}) = \bigoplus\nolimits_{i_0 < \ldots < i_p,\ NEG(\vec{e}) \subset \{i_0, \ldots, i_p\}} R \cdot T_0^{e_0} \ldots T_n^{e_n}$$

In other words, the terms corresponding to $i_0 < \ldots < i_p$ such that $NEG(\vec{e})$ is not contained in $\{i_0 \ldots i_p\}$ are zero. The differential of the complex $\check{\mathcal{C}}^\bullet(\vec{e})$ is still given by the exact same formula as above.

Suppose that $NEG(\vec{e}) = \{0, \ldots, n\}$, i.e., that all exponents e_i are negative. In this case the complex $\check{\mathcal{C}}^\bullet(\vec{e})$ has only one term, namely $\check{\mathcal{C}}^n(\vec{e}) = R \cdot \frac{1}{T_0^{-e_0} \ldots T_n^{-e_n}}$. Hence in this case

$$H^q(\check{\mathcal{C}}^\bullet(\vec{e})) = \begin{cases} R \cdot \frac{1}{T_0^{-e_0} \ldots T_n^{-e_n}} & \text{if} \quad q = n \\ 0 & \text{if} \quad \text{else} \end{cases}$$

The direct sum of all of these terms clearly gives the value

$$\left(\frac{1}{T_0 \ldots T_n} R[\frac{1}{T_0}, \ldots, \frac{1}{T_n}]\right)_d$$

in degree n as stated in the lemma. Moreover these terms do not contribute to cohomology in other degrees (also in accordance with the statement of the lemma).

Assume $NEG(\vec{e}) = \emptyset$. In this case the complex $\check{\mathcal{C}}^\bullet(\vec{e})$ has a summand R corresponding to all $i_0 < \ldots < i_p$. Let us compare the complex $\check{\mathcal{C}}^\bullet(\vec{e})$ to another complex. Namely, consider the affine open covering

$$\mathcal{V} : \mathrm{Spec}(R) = \bigcup\nolimits_{i \in \{0,\ldots,n\}} V_i$$

where $V_i = \mathrm{Spec}(R)$ for all i. Consider the alternating Čech complex

$$\check{\mathcal{C}}^\bullet_{ord}(\mathcal{V}, \mathcal{O}_{\mathrm{Spec}(R)})$$

By the same reasoning as above this computes the cohomology of the structure sheaf on $\mathrm{Spec}(R)$. Hence we see that $H^p(\check{\mathcal{C}}^\bullet_{ord}(\mathcal{V}, \mathcal{O}_{\mathrm{Spec}(R)})) = R$ if $p = 0$ and is 0 whenever $p > 0$. For these facts, see Lemma 01X9 and its proof. Note that also $\check{\mathcal{C}}^\bullet_{ord}(\mathcal{V}, \mathcal{O}_{\mathrm{Spec}(R)})$ has a summand R for every $i_0 < \ldots < i_p$ and has exactly the same differential as $\check{\mathcal{C}}^\bullet(\vec{e})$. In other words these complexes are isomorphic complexes and hence have the same cohomology. We conclude that

$$H^q(\check{\mathcal{C}}^\bullet(\vec{e})) = \begin{cases} R \cdot T_0^{e_0} \ldots T_n^{e_n} & \text{if} \quad q = 0 \\ 0 & \text{if} \quad \text{else} \end{cases}$$

in the case that $NEG(\vec{e}) = \emptyset$. The direct sum of all of these terms clearly gives the value

$$(R[T_0, \ldots, T_n])_d$$

in degree 0 as stated in the lemma. Moreover these terms do not contribute to cohomology in other degrees (also in accordance with the statement of the lemma).

To finish the proof of the lemma we have to show that the complexes $\check{\mathcal{C}}^\bullet(\vec{e})$ are acyclic when $NEG(\vec{e})$ is neither empty nor equal to $\{0, \ldots, n\}$. Pick an index $i_{\mathrm{fix}} \notin NEG(\vec{e})$ (such an index exists). Consider the map

$$h : \check{\mathcal{C}}^{p+1}(\vec{e}) \to \check{\mathcal{C}}^p(\vec{e})$$

given by the rule that for $i_0 < \ldots < i_p$ we have

$$h(s)_{i_0 \ldots i_p} = \begin{cases} 0 & \text{if} \quad p \notin \{0, \ldots, n-1\} \\ 0 & \text{if} \quad i_{\mathrm{fix}} \in \{i_0, \ldots, i_p\} \\ s_{i_{\mathrm{fix}} i_0 \ldots i_p} & \text{if} \quad i_{\mathrm{fix}} < i_0 \\ (-1)^a s_{i_0 \ldots i_{a-1} i_{\mathrm{fix}} i_a \ldots i_p} & \text{if} \quad i_{a-1} < i_{\mathrm{fix}} < i_a \\ (-1)^{p+1} s_{i_0 \ldots i_p i_{\mathrm{fix}}} & \text{if} \quad i_p < i_{\mathrm{fix}} \end{cases}$$

Please compare with the proof of Lemma 01X9. This makes sense because we have

$$NEG(\vec{e}) \subset \{i_0, \ldots, i_p\} \Leftrightarrow NEG(\vec{e}) \subset \{i_{\mathrm{fix}}, i_0, \ldots, i_p\}$$

The exact same (combinatorial) computation[1] as in the proof of Lemma 01X9 shows that

$$(hd + dh)(s)_{i_0 \ldots i_p} = s_{i_0 \ldots i_p}$$

Hence we see that the identity map of the complex $\vec{\mathcal{C}}^\bullet(\vec{e})$ is homotopic to zero which implies that it is acyclic. □

[1]For example, suppose that $i_0 < \ldots < i_p$ is such that $i_{\text{fix}} \notin \{i_0, \ldots, i_p\}$ and that $i_{a-1} < i_{\text{fix}} < i_a$ for some $1 \leq a \leq p$. Then we have

$$\begin{aligned}
&(dh + hd)(s)_{i_0 \ldots i_p} \\
&= \sum_{j=0}^{p} (-1)^j h(s)_{i_0 \ldots \hat{i}_j \ldots i_p} + (-1)^a d(s)_{i_0 \ldots i_{a-1} i_{\text{fix}} i_a \ldots i_p} \\
&= \sum_{j=0}^{a-1} (-1)^{j+a-1} s_{i_0 \ldots \hat{i}_j \ldots i_{a-1} i_{\text{fix}} i_a \ldots i_p} + \sum_{j=a}^{p} (-1)^{j+a} s_{i_0 \ldots i_{a-1} i_{\text{fix}} i_a \ldots \hat{i}_j \ldots i_p} + \\
&\sum_{j=0}^{a-1} (-1)^{a+j} s_{i_0 \ldots \hat{i}_j \ldots i_{a-1} i_{\text{fix}} i_a \ldots i_p} + (-1)^{2a} s_{i_0 \ldots i_p} + \sum_{j=a}^{p} (-1)^{a+j+1} s_{i_0 \ldots i_{a-1} i_{\text{fix}} i_a \ldots \hat{i}_j \ldots i_p} \\
&= s_{i_0 \ldots i_p}
\end{aligned}$$

as desired. The other cases are similar.